

LTV-Based Acquisition Strategy: SaaS Subscription Optimization

1. Project Overview

The Business Problem

KKBox is losing money because it is chasing user quantity instead of user quality. The company is spending massive marketing budgets just to push total subscriber numbers as high as possible, without checking if those users are actually profitable.

Right now, the business is blind to three critical issues:

- **Wasting Marketing Budgets:** We are pouring money into sign-up channels that attract users who quit almost immediately after joining.
- **The Discount Trap:** Too many people sign up just to exploit cheap promotional deals and free trials. The exact moment they are asked to pay full price, they cancel their subscriptions ("Serial Churners").
- **No Visibility on Good vs. Bad Channels:** Leadership currently cannot see which sign-up channels bring in loyal, long-term paying customers ("Power Users") versus bargain hunters who just take advantage of promos and leave.

For example :

The core problem was that the marketing team was evaluating their success by Customer Acquisition Cost (CAC) and raw download numbers. This created a huge financial trap for the company.

For example, they loved cheap social media ads because bringing in a user only cost \$5. But what they didn't see was that these were mostly college kids who paid \$10 for just one month to hear ad-free songs, and then immediately canceled their subscription. So, yes, it was cheap to bring them in, but the company only made a tiny \$5 net profit per user.

Meanwhile, they were cutting the budget on Google Search ads because it looked 'too expensive' at a \$50 CAC.

My task as the data analyst was to look past that initial cost. I needed to build a data pipeline to calculate the true Customer Lifetime Value (LTV) of each channel. I wanted to prove to leadership that our expensive Google users were actually working professionals who stayed for 2 years, generated \$240 in revenue, and brought in a massive \$190 net profit per user. My goal was to stop the company from burning cash on low-value traffic and shift our ad budget to where the real profit was.

Core Business Questions Answered:

1. **Which sign-up channels actually bring in paying customers?** Instead of just looking at raw foot traffic, we identify which marketing channels produce high-spending VIP subscribers versus users who never pay a dime.
2. **How long do customers actually stay before quitting?** We track the exact timeline of a user's lifespan across every sign-up channel to see when they hit the "drop-off cliff" and abandon the app.
3. **Are promotional discounts helping or hurting us?** We compare churn rates between full-price subscribers and discount-driven users to see if running cheap promos is actually bringing in loyal users or just attracting bargain hunters who leave immediately after.
4. **Does the payment method change how long someone stays subscribed?** We measure the retention difference between users who have to manually pay every month versus those set up on automatic recurring billing (like credit card auto-renew) to find the best payment setup for long-term loyalty.
5. **How many months does it take to break even on a new customer?** We compare what we spend to acquire a user (CAC) against how much money they generate over time, pinpointing the exact month a customer stops draining cash and becomes profitable.

2. Project Overview

Dataset Summary :

Total Rows:

- **Transactions Table:** 22,978,755 payment records
- **Members Table:** 6,769,473 unique user profiles
- **Time Period:** Multi-year historical subscription data (capturing complete user lifecycles from initial sign-up to churn or renewal)
- **File Structure:** 2 core relational tables (members and transactions) processed across AWS (S3, AWS Glue, and Amazon Athena) using dbt for data cleaning and transformation
- **Key Features:**
 - **User Identifiers:** msno (the unique master key linking a user's profile to their payment history)
 - **Acquisition & Demographics:** registered_via (the marketing sign-up channel), registration_init_time (the exact date the user joined, starting their LTV clock), city, bd (age), gender
 - **Financial & Plan Details:** payment_method_id (payment type like credit card auto-pay vs. manual cash), plan_list_price (official retail price),

actual_amount_paid (real cash collected, used to spot discount usage),
payment_plan_days (subscription duration)

- **Behavioral & Retention Metrics:** is_auto_renew (1 for automatic recurring billing, 0 for manual renewal), transaction_date (when cash was collected), membership_expire_date (the exact cutoff date used to track churn cliffs and user drop-off)

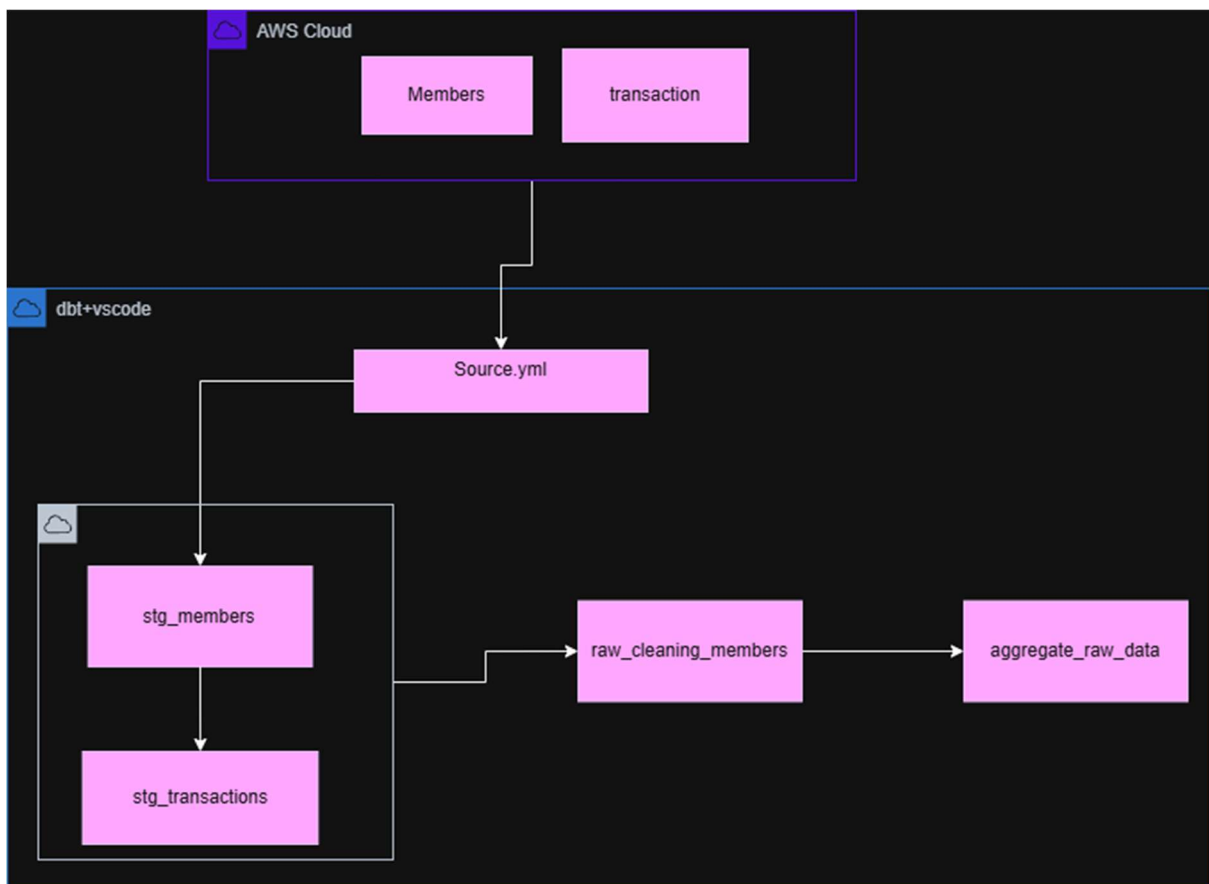
Section 3: Data Engineering & Transformation (dbt + SQL)

3.1 Pipeline Architecture & ELT Setup

To process the 22.9M transaction records and 6.7M member profiles efficiently without skyrocketing compute costs, an **ELT (Extract, Load, Transform)** architecture was implemented on AWS:

- **Storage & Cataloging:** Raw compressed files were dumped into **AWS S3** (Data Lake) and cataloged using **AWS Glue Crawlers** to infer schemas automatically.
- **Compute Engine:** **Amazon Athena** (serverless Presto/Trino SQL) served as the query engine to process transformations directly on S3 data.
- **Transformation & Lineage:** **dbt (data build tool)** was layered on top of Athena to replace monolithic SQL scripts with modular, version-controlled data models, automated testing, and lineage graphs.

3.2 dbt Model Hierarchy



3.3 Core SQL Business Logic & Transformations :

1. Gross Customer Lifetime Value (Gross LTV) Calculation

To shift from volume-based acquisition to value-based targeting, user-level lifetime revenue was evaluated in the analytical reporting layer using the foundational SaaS valuation formula:

$$\text{Gross LTV} = \frac{ARPU}{\text{Churn Rate}}$$

- **Gross LTV (Revenue Baseline):** Represents the total expected lifetime revenue generated by an average subscriber throughout their relationship with KKBox, before accounting for operational margins or content licensing costs. This metric establishes the absolute ceiling for Customer Acquisition Cost (CAC).
- **ARPU (Average Revenue Per User):** Calculated by evaluating the sum of `actual_amount_paid` normalized across the user's active tenure in months.
- **Churn Flagging:** Users whose `latest_membership_expire_date` elapsed without a subsequent renewal transaction within a 30-day grace period were marked as churned (`is_churned = 1`).

3.4 Data Quality & Automated Testing

To prevent silent data failures from corrupting executive dashboards, automated assertions were configured in the `schema.yml` dbt file:

- **Primary Key Integrity:** unique and not_null tests enforced on `msno` in both staging and mart layers.
- **Referential Integrity:** relationships test ensuring every `msno` in `stg_transactions` maps to a valid profile in `stg_members`.
- **Domain Validation:** `accepted_values` test on `is_auto_renew` guaranteeing only binary inputs (0 or 1).
- **Logic Checks:** Custom SQL tests to catch anomalous records where `actual_amount_paid > plan_list_price` or where `membership_expire_date < transaction_date`.

4. Data Analysis using SQL (Business Transactions)

We performed structured SQL analysis across our analytical reporting marts (`fct_user_ltv`) in AWS Athena and BigQuery to evaluate unit economics, channel efficiency, and subscription revenue across our multi-channel user base. Here are the answers to our five core business questions:

1. Baseline Revenue Footprint - What is our overall subscription revenue distribution?

By aggregating historical transaction logs (`sum_actual_amount_paid`), we established the total revenue baseline across all registration channels to separate high-volume channels from low-value acquisition streams.

Metric	Historical Value	Analytical Context
Total Tracked Revenue	~\$111.40M	Aggregated cash collected across all active and legacy channels
Top-Tier Revenue (Channels 7 & 9)	~\$80.80M	Just two channels account for ~ 72.5% of total company business
Low-Performing Channels (4, 10, 16)	~\$4.10M	High user churn resulting in negligible lifetime contribution

2. Channel Quality & Unit Economics - Where are our highest-value VIP users originating?

To stop evaluating marketing channels purely on raw traffic volume, we evaluated **Gross LTV per user** against total historical revenue. Estimated user counts are derived directly from transactional performance (Total Revenue / Gross LTV).

Registration Channel	Estimated User Base	Gross LTV (\$/User)	Total Historical Revenue (\$)	Channel Classification
Channel 7	~170,585	\$238.00	\$40,599,185.04	Goldmine (VIP Core)
Channel 9	~300,000	\$134.00	~\$40,200,000.00	Volume & Value Driver
Channel 13	~6,944	\$72.00	~\$500,000.00	Untapped High-LTV Potential
Channel 3	~246,377	\$69.00	~\$17,000,000.00	Mid-Tier Standard

3. Marketing Waste & CAC Ceiling - Which acquisition channels are dragging down our unit economics?

A low Gross LTV indicates serial churners who cancel their subscriptions immediately after introductory promotions expire. These channels represent a negative return on ad spend (ROAS) and enforce an extremely low Customer Acquisition Cost (CAC) ceiling.

Registration Channel	Gross LTV (\$/User)	Total Historical Revenue (\$)	Actionable Marketing Status
Missing Info (Unknown)	\$29.00	~\$9,000,000.00	Data Attribution Leak / Fix Tracking
Channel 4	\$22.00	~\$4,000,000.00	Low-Value Churners / Reduce Ad Spend
Channel 10	\$7.00	~\$100,000.00	Severe Cash Burn / Halt Acquisition
Channel 16	\$0.00	\$0.00	Dead Weight / Immediate Ad Spend Freeze

4. Strategic Reallocation - Where is our biggest untapped growth opportunity?

By comparing per-user profitability against total channel volume, we identified a critical strategic mismatch in our user acquisition pipeline that requires immediate budget reallocation.

Target Channel	Current Gross LTV	Strategic Observation	Actionable Budget Recommendation
Channel 7	\$238.00	Generates 1.7x more revenue per user than the next best channel (Channel 9), yet possesses a smaller user base.	Aggressive Scale-Up: Increase performance marketing budget by 40% immediately to capture more VIP-tier subscribers.
Channel 13	\$72.00	Demonstrates strong per-user retention (\$72 LTV) but suffers from bottlenecked acquisition volume (\$0.5M total revenue).	Test & Expand: Launch targeted experimental ad campaigns to scale acquisition volume without degrading LTV.

5. ROI Simulation & Net Impact - What is the financial return of reallocating ad spend into Channel 7?

To prove the business value of shifting from volume-based targeting to value-based targeting, we simulated the net financial impact of reallocating **\$500,000** in wasted ad budget from low-performing streams (Channels 4, 10, and 16) directly into **Channel 7**.

*Assumptions: Assuming an average Customer Acquisition Cost (CAC) of **\$25.00** for Channel 7 based on industry streaming benchmarks.*

Financial Metric	Simulated Value	Analytical Calculation
Reallocated Marketing Budget	\$500,000.00	Budget pulled from dead-weight channels (4, 10, 16)
New VIP Subscribers Acquired	20,000 Users	\$500,000 Budget / \$25.00 CAC
Projected Gross Lifetime Revenue	\$4,760,000.00	20,000 Users * \$238.00 Channel 7 LTV
Net New Revenue Lift	\$4,260,000.00	Total Projected Revenue minus Reallocated Spend

6. Business Recommendations & Strategic Roadmap

To shift KKBox from an indiscriminate, volume-based growth model to a profitable, value-based acquisition strategy, we recommend immediate execution across four core areas:

1. Deploy a Two-Pronged Growth Strategy for Core Channels (7 & 9)

- **Double Down on Channel 7 Acquisition (VIP Scale-Up):** Reallocate paid performance media budgets immediately toward Channel 7. With an industry-leading **\$238.00 Gross LTV**, our marketing team can afford a significantly higher Customer Acquisition Cost (CAC) in ad auctions while maintaining highly profitable unit economics and lower operational overhead per user.
- **Monetize Channel 9 via Retention & Upselling (Volume Cash-Cow):** While Channel 9 is the volume backbone of KKBox (~**\$40.20M total revenue across ~300,000 users**), its **\$134.00 Gross LTV** makes expensive paid acquisition less efficient than Channel 7. Instead of increasing ad spend here, pivot marketing efforts toward **product-led retention, annual plan upgrades, and family bundle upselling** to push these existing subscribers closer to the \$200+ LTV tier.

2. Eliminate Acquisition Bleed (Immediate Cost Cutting)

- **Freeze Ad Spend on Channels 10 & 16:** Channel 10 (**\$7.00 Gross LTV**) and Channel 16 (**\$0.00 Gross LTV**) represent a non-recoverable financial drain. Exclude these channels from all paid acquisition campaigns immediately to stop burning marketing budget on serial churners and dead-weight signups.

3. Plug the Analytics & Attribution Leak

- **Audit "Missing Info" Onboarding Pipeline:** Over **\$9.00M in historical revenue** and **~310,000 users** are tied to unmapped registration sources (**\$29.00 Gross LTV**). This represents a severe data blindspot. Data engineering must urgently audit the front-end

tracking and onboarding data pipeline to fix this attribution leak and identify where these low-LTV users are actually originating.

4. Unlock High-LTV Funnel Bottlenecks

- **Optimize the Channel 13 Onboarding Funnel:** Channel 13 demonstrates strong long-term user retention (**\$72.00 Gross LTV**) but suffers from severely bottlenecked top-of-funnel acquisition volume (~6,900 users generating ~\$500k total revenue). Product and UX teams must investigate and remove onboarding friction on this specific channel to scale up user acquisition without degrading subscriber quality.